

The log-squared weight is sharp for the borderline logarithmic heavy tail

Candidate full answer to Example 2.28 of arXiv:0807.3112

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Status. Candidate full solution, pending expert review. For the probability density in the source, the weight $1 + x^2 \log^2(2 + |x|)$ is exactly the asymptotic threshold for a weighted Poincaré inequality. In particular, the exponent 2 cannot be lowered, and no eventual $o(x^2 \log^2 x)$ weight can replace it.

Source. P. Cattiaux, N. Gozlan, A. Guillin, and C. Roberto, *Functional inequalities for heavy tails distributions and application to isoperimetry*, arXiv:0807.3112, Example 2.28 on printed page 12.¹ After proving

$$\mathrm{Var}_{\mu_q}(f) \leq C_q \int |f'(x)|^2 (1 + x^2 \log^2(2 + |x|)) d\mu_q(x),$$

the authors ask whether this weight is correct. The same page recalls the one-dimensional Muckenhoupt product that resolves the question.

¹P. Cattiaux, N. Gozlan, A. Guillin, and C. Roberto, arXiv:0807.3112 (2008).

Example 2.28. Let $q > 1$ and define

$$d\mu(x) = (1/Z_q) ((2 + |x|) \log^q(2 + |x|))^{-1} dx = V_q^{-1}(x) dx \quad x \in \mathbb{R}.$$

The function V_q is convex but V_q^γ is no more convex for $\gamma < 1$ (hence $\kappa = -\infty$). We may choose $W(x) = (2 + |x|)^2 \log^a(2 + |x|)$ (at least far from 0), which is a ϕ -Lyapunov function for $\phi(u) = \log^{a-1}(2 + |u|)$ provided $q > a > 1$ (details are left to the reader). We thus get a weighted inequality

$$(2.29) \quad \text{Var}_\mu(g) \leq C \int |\nabla g(x)|^2 (1 + x^2 \log^2(2 + |x|)) d\mu(x).$$

Unfortunately we do not know whether the weight is correct in this situation. The usual choice g behaving like $\sqrt{(2 + |x|) \log^q(2 + |x|)}$ on $(-R, R)$ furnishes a variance behaving like R but the right hand side behaves like $R \log^2 R$.

Figure 1: Example 2.28 and its open weight question in the archived source PDF, printed page 12.

1 Proof intuition

In one dimension the issue is decided by a single tail-energy product: the mass beyond y is multiplied by the reciprocal weighted energy needed to climb from the median to y . For the present law these two factors have opposite logarithmic powers, and their product stays bounded exactly when the weight supplies two powers of $\log x$. The Hardy criterion proves sufficiency, while the associated one-sided equilibrium potentials are matching test functions and therefore prove necessity.

2 Exact criterion and answer

Fix $q > 1$, put $L(x) = \log(2 + x)$ for $x \geq 0$, and let

$$p_q(x) = \frac{1}{Z_q(2 + |x|) \log^q(2 + |x|)}, \quad d\mu_q(x) = p_q(x) dx.$$

For a positive even continuous weight W , define

$$\begin{aligned} T_q(y) &= \mu_q([y, \infty)), \\ I_W(y) &= \int_0^y \frac{dx}{W(x)p_q(x)}, \\ B_W &= \sup_{y>0} T_q(y) I_W(y). \end{aligned}$$

Let C_W be the least constant in

$$\text{Var}_{\mu_q}(f) \leq C_W \int_{\mathbb{R}} |f'(x)|^2 W(x) d\mu_q(x) \quad (1)$$

on the natural weighted Sobolev form domain.

Theorem 1 (sharp criterion and threshold). *The weighted Poincaré inequality (1) holds if and only if $B_W < \infty$, and its best constant satisfies*

$$\frac{1}{2} B_W \leq C_W \leq 4 B_W. \quad (2)$$

Consequently:

1. for $W_\beta(x) = 1 + x^2 \log^\beta(2 + |x|)$, inequality (1) holds if and only if $\beta \geq 2$;
2. if $W(x) = o(x^2 \log^2(2 + x))$ as $x \rightarrow \infty$, then (1) fails;
3. if $W(x) \geq cx^2 \log^2(2 + x)$ eventually for some $c > 0$, then (1) holds.

Thus the source's log-squared weight is sharp in both the logarithmic-power scale and the stronger eventual-comparison sense.

3 Proof

The upper inequality in (2) is the one-dimensional weighted Hardy theorem used immediately before Example 2.28 in the source. On $[0, \infty)$ it says

$$\int_0^\infty (f(x) - f(0))^2 d\mu_q(x) \leq 4B_W \int_0^\infty |f'(x)|^2 W(x) d\mu_q(x).$$

Apply it on both half-lines and use $\text{Var}_{\mu_q}(f) \leq \int (f - f(0))^2 d\mu_q$ to obtain $C_W \leq 4B_W$.

For completeness, the converse and the obstruction can be seen directly. For $y > 0$ set

$$h_y(x) = \begin{cases} 0, & x \leq 0, \\ \int_0^{\min\{x,y\}} \frac{ds}{W(s)p_q(s)}, & x > 0. \end{cases}$$

Then

$$\int |h'_y|^2 W d\mu_q = I_W(y), \quad \int h_y^2 d\mu_q \geq T_q(y) I_W(y)^2. \quad (3)$$

Because h_y is supported on a half-line of μ_q -mass $1/2$, Cauchy–Schwarz gives

$$\left(\int h_y d\mu_q \right)^2 \leq \frac{1}{2} \int h_y^2 d\mu_q,$$

so $\text{Var}_{\mu_q}(h_y) \geq \frac{1}{2} \int h_y^2 d\mu_q$. Substitution in (1) and (3) yields $C_W \geq \frac{1}{2} T_q(y) I_W(y)$. Taking the supremum proves the lower inequality in (2). Local smoothing at 0 and y , while retaining the two constant tails, gives the same conclusion in a smooth formulation of (1).

It remains only to calculate the product. The tail is exact:

$$T_q(y) = \frac{1}{Z_q(q-1)} L(y)^{1-q}. \quad (4)$$

For W_β ,

$$\begin{aligned} I_{W_\beta}(y) &= Z_q \int_0^y \frac{(2+x)L(x)^q}{1+x^2L(x)^\beta} dx \\ &\sim Z_q \int_0^y \frac{L(x)^{q-\beta}}{x} dx. \end{aligned} \quad (5)$$

Hence, when $\beta < q+1$,

$$I_{W_\beta}(y) \sim \frac{Z_q}{q-\beta+1} L(y)^{q-\beta+1}, \quad T_q(y) I_{W_\beta}(y) \sim \frac{L(y)^{2-\beta}}{(q-1)(q-\beta+1)}. \quad (6)$$

At $\beta = q+1$, the integral in (5) is asymptotic to $Z_q \log L(y)$; for $\beta > q+1$ it converges. It follows that the product is bounded exactly when $\beta \geq 2$. At the endpoint,

$$T_q(y) I_{W_2}(y) \longrightarrow \frac{1}{(q-1)^2}, \quad (7)$$

which also displays the genuine criticality of the exponent 2.

Finally suppose $W(x) = o(x^2 L(x)^2)$. Given $M > 0$, for all sufficiently large x ,

$$\frac{1}{W(x)p_q(x)} \geq M Z_q \frac{(2+x)L(x)^{q-2}}{x^2}.$$

After integration, (4) gives $\liminf_{y \rightarrow \infty} T_q(y) I_W(y) \geq c_q M$. As M is arbitrary, $B_W = \infty$, proving failure. Conversely, an eventual lower bound by a positive multiple of $x^2 L(x)^2$, together with positivity and continuity on compact intervals, makes B_W finite by comparison with the endpoint case. This completes the proof.

4 What the source’s test missed

The source tests a truncated square root of the reciprocal density; for this borderline tail it leaves a spurious factor $\log^2 R$. The correct near-extremizers are the truncated Hardy potentials h_y above. For a putative power W_β their Rayleigh quotient obeys

$$\frac{\text{Var}_{\mu_q}(h_y)}{\int |h'_y|^2 W_\beta d\mu_q} \geq \frac{1}{2} T_q(y) I_{W_\beta}(y) \asymp \log^{2-\beta} y,$$

which diverges precisely below the exponent 2.

Scope and novelty. The conclusion is a full order-sharp answer to the source’s informal question, not a claim that one pointwise-minimal weight exists among arbitrary oscillatory weights. Such weights are classified exactly by B_W . The proof is a direct specialization of Muckenhoupt’s classical weighted Hardy theorem (*Studia Math.* **44** (1972), 31–38), which the source itself invokes on the same printed page. Bounded exact-phrase, title/citation, density, and weighted-Poincaré searches on 13 August 2026 did not locate a later paper explicitly recording this calculation. No priority or independent novelty is claimed.

Human-review recommendation. Check the full-line lower bound using the one-sided h_y , the exponent in (5), and the endpoint limit (7). These are the only substantive points; normalizing constants cancel from the threshold computation.